Day 2.

Block 1

Aptitude (Profit & Loss).

1) CP = 400
SP = 480 (SP > CP)

SP − CP = 480 − 400
Profit = 80

Profit % = $\frac{SP-CP}{CP} \times 100$

$= \frac{80}{400} \times 100$

$\boxed{\text{Profit\%} = 20\%}$

2) CP = 1500
SP = 1200 (SP < CP) → loss

loss = CP − SP
= 1500 − 1200
Loss = 300.

$\frac{300}{1500} \times 100$

Loss Percentage = 20%.

3) Let CP = 100 Rupees

Marked Price = 100 + 20% = 120
Discount = 10% of 120 = 12/-

$\frac{10}{100} \times 120$

SP = 120 − 12 = 108 Rupees
Profit = 108 − 100 ⇒ 8
Profit % = $\frac{8}{100} \times 100$
= 8 %.

4)

Cost of 5 Pens = ₹100

$$P\% = \frac{SP - CP}{CP} \times 100.$$

25

$\frac{125}{-5}$

$$25 = \frac{SP - 100}{100}$$

$$25 = SP - 100$$

$$SP = 125$$

∴ 25 Rupees per pen

5)

SP = ₹900 loss % = 10

$\frac{SP}{CP} = \frac{90}{100}$

$900 \times \frac{10}{100}$

90%

SP = 90% of CP

$\frac{900}{CP} = \frac{90}{100}$

$$CP = \frac{SP}{0.9} = \frac{900}{0.9} = 1000$$

90000 = CP(90)

$CP = \frac{90000}{90}$

CP = 10000

∴ ₹1,000

Block - 2

Verbal.

Reading passage:

1) Technologies like cloud computing and artificial intelligence are influencing software development.

2) scalable system are capable of handling millions of users simultaneously

3) scalable in software system mean the ability of a system handle large amount of data or workload without slowing down.

Grammar correction:

1) He don't like programming

He doesn't like programming

2) She have submitted the assignment

She has submitted the assignment

3) They was discussing the project

They were discussing the project.

Block 3.

Final year project.

Hello, I would like to explain my final year project idea.

My Project title is Hospital Management system. This project focuses on improving the efficiency of hospital operations by digitalizing key processes.

The main problem is that many small and medium-scale hospital, still rely on manual record-keeping. This often leads to data loss, delays in accessing patient information and difficulties in managing appointments and billing.

To address this issue, I am planning to develop a web-based application which makes the entire process easier.

For this project, I am going to use MERN stack.

My role in this project is, responsible for designing the user interface, creating backend APIs and managing the database.

This is going to a solo project, am going to build the entire system by my own.

As am not started working on this project for now, I haven't faced any challenges.

But one of the major expected challenges is integrating multiple modules efficiently so that the system performs smoothly even with large amounts of data.

Overall, this project will help me to gain hands-on experience in full-stack development and gives me an opportunity to solve real-world problems in the healthcare domain.